

Healthcare Claims Data Analysis - Answers

Data Files Used

- **STONYBRK_20240531_HEADER.csv**: 388 claims, 43 columns
 - **STONYBRK_20240531_LINE.csv**: 520 service lines, 28 columns
 - **STONYBRK_20240531_CODE.csv**: 1,536 diagnosis codes, 9 columns
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Part 1: Data Exploration

Key Observations

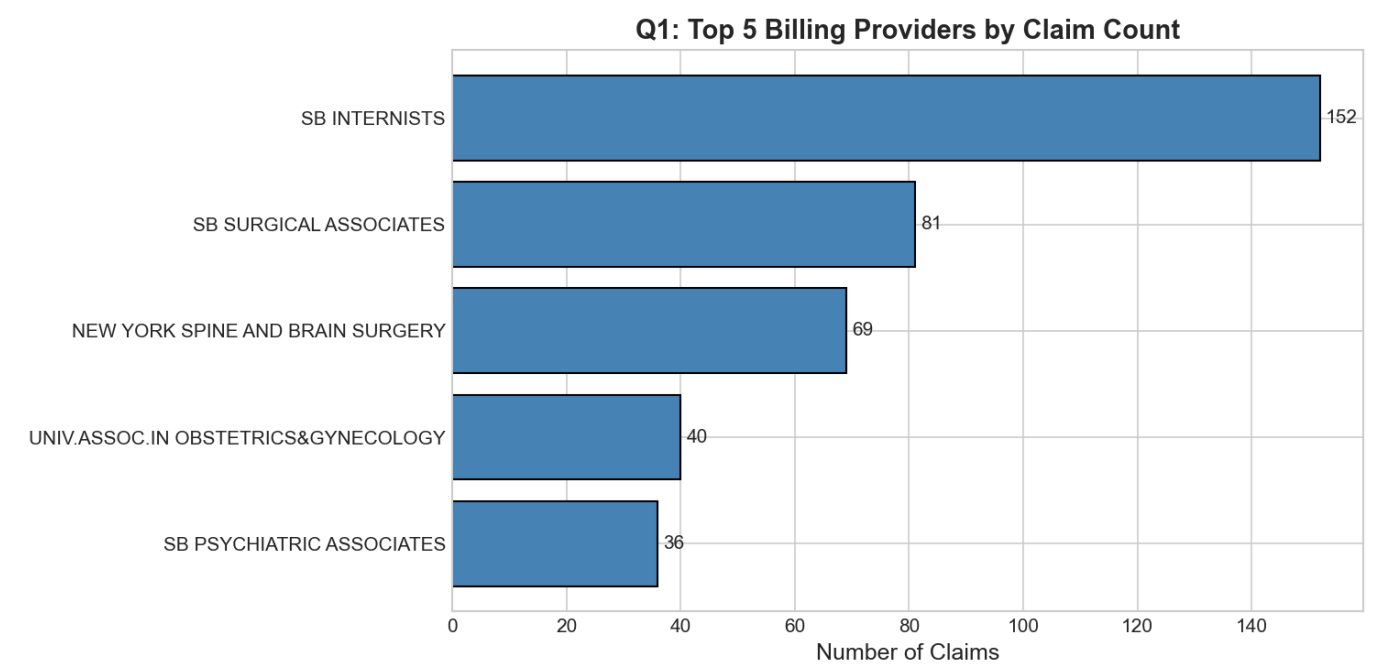
Metric	Value
Unique Claims	388
Date Range	Sept 25, 2023 - May 29, 2024
Avg Service Lines per Claim	1.34
Avg Diagnosis Codes per Claim	3.96

Observations:

- The dataset spans approximately 8 months of claims data
 - Most claims have only 1-2 service lines, indicating relatively straightforward billing
 - Claims have nearly 4 diagnosis codes on average, suggesting moderate case complexity
 - The relational structure links claims (HEADER) to procedures (LINE) and diagnoses (CODE) through **ProspectiveClaimId**
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Part 2: Relational Data Analysis

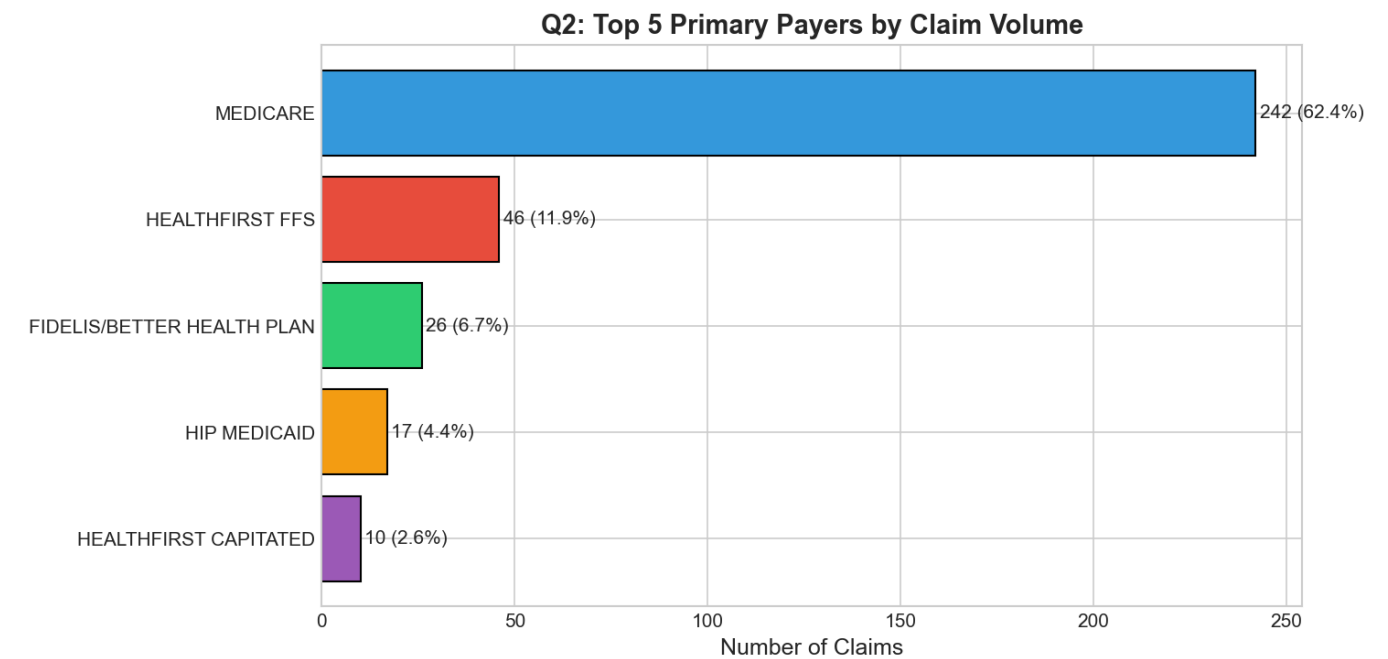
Question 1: Top 5 Billing Providers by Claim Count



Provider Name	NPI	Claim Count
SB INTERNISTS	1821035601	152
SB SURGICAL ASSOCIATES	1063468536	81
NEW YORK SPINE AND BRAIN SURGERY	1154376309	69
UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	1538114723	40
SB PSYCHIATRIC ASSOCIATES	1437105905	36

Interpretation: SB Internists dominates the claim volume with 152 claims (39% of total), which is typical for an internal medicine group handling general patient care. Surgical Associates and Spine & Brain Surgery follow, indicating significant procedural volume at the hospital.

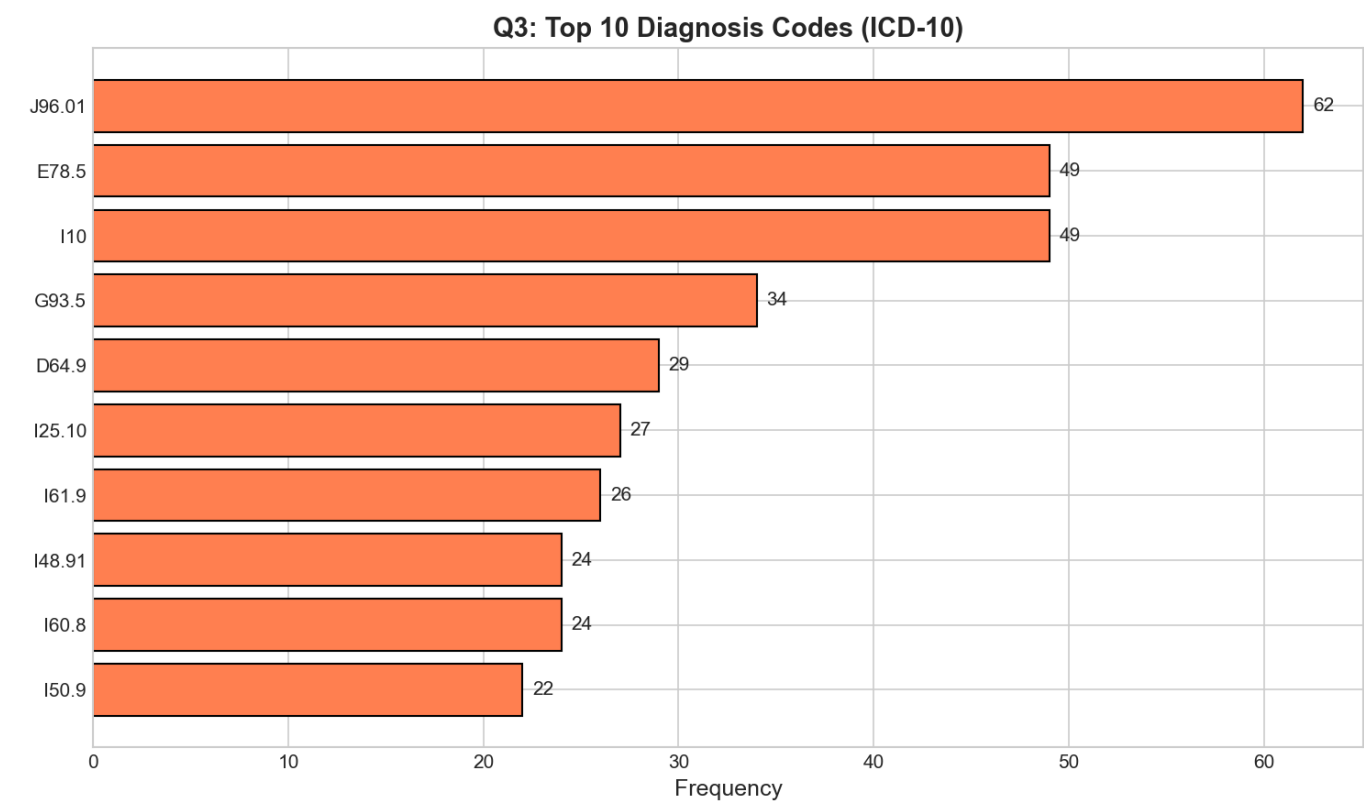
Question 2: Payer Mix Analysis



Payer	Claims	Percentage
MEDICARE	242	62.4%
HEALTHFIRST FFS	46	11.9%
FIDELIS/BETTER HEALTH PLAN	26	6.7%
HIP MEDICAID	17	4.4%
HEALTHFIRST CAPITATED	10	2.6%

Interpretation: Medicare is the dominant payer at 62.4% of claims, which is common for academic medical centers serving older populations. Medicaid-related payers (Healthfirst, Fidelis, HIP) collectively account for about 25% of claims, reflecting the hospital's role in serving underserved populations. Commercial payers are notably underrepresented in this sample.

Question 3: Top 10 Diagnosis Codes

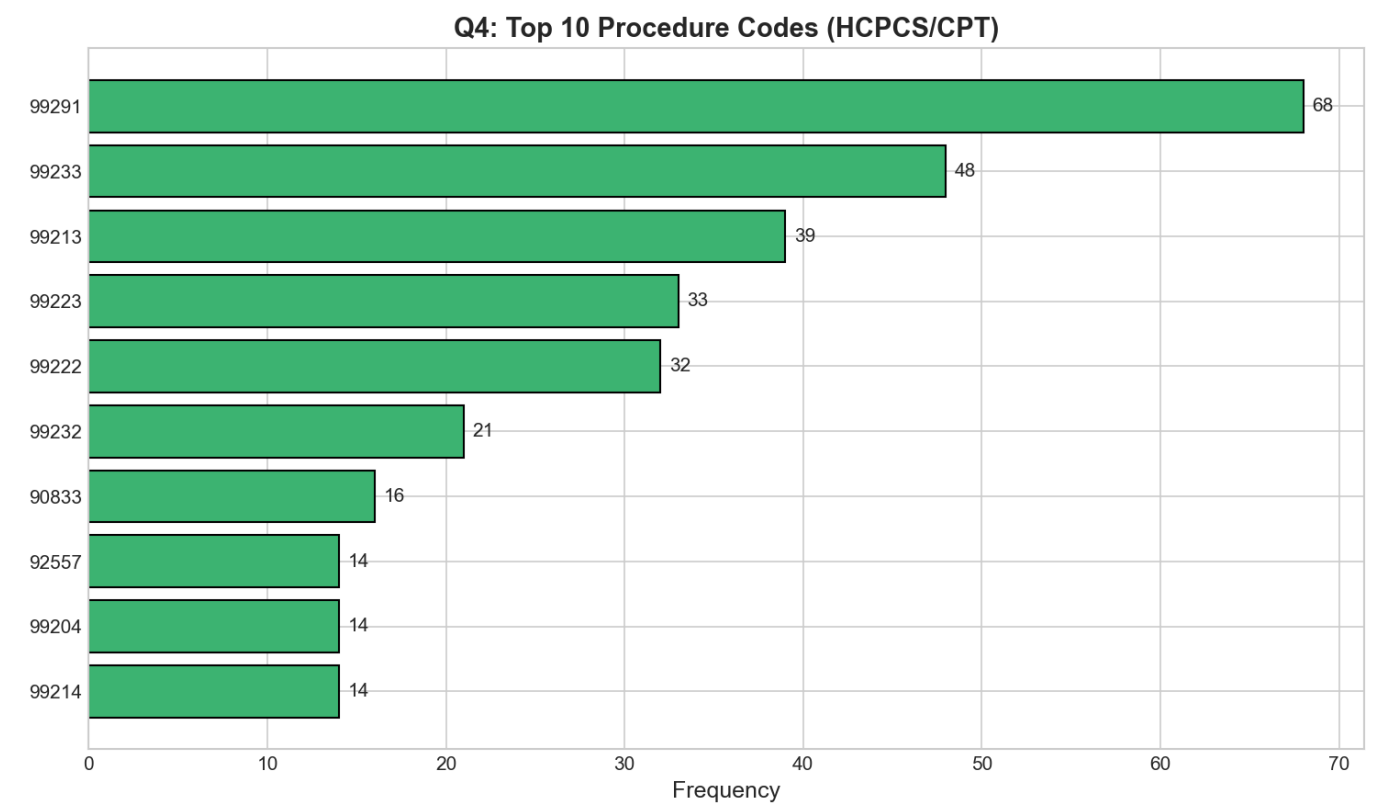


ICD-10 Code	Description	Frequency
J96.01	Acute respiratory failure with hypoxia	62
E78.5	Hyperlipidemia, unspecified	49
I10	Essential (primary) hypertension	49
G93.5	Compression of brain	34
D64.9	Anemia, unspecified	29
I25.10	Atherosclerotic heart disease	27
I61.9	Nontraumatic intracerebral hemorrhage, unspecified	26
I48.91	Atrial fibrillation, unspecified	24
I60.8	Other nontraumatic subarachnoid hemorrhage	24
I50.9	Heart failure, unspecified	22

Interpretation: The top diagnosis codes reveal a patient population with:

- **Critical illness:** Acute respiratory failure (J96.01) is the most common code
- **Cardiovascular disease:** Hypertension, heart disease, heart failure, and atrial fibrillation are prevalent
- **Neurological conditions:** Brain compression, intracerebral hemorrhage, and subarachnoid hemorrhage appear frequently, consistent with the Spine & Brain Surgery provider presence
- **Chronic conditions:** Hyperlipidemia and anemia are common comorbidities

Question 4: Top 10 Procedure Codes



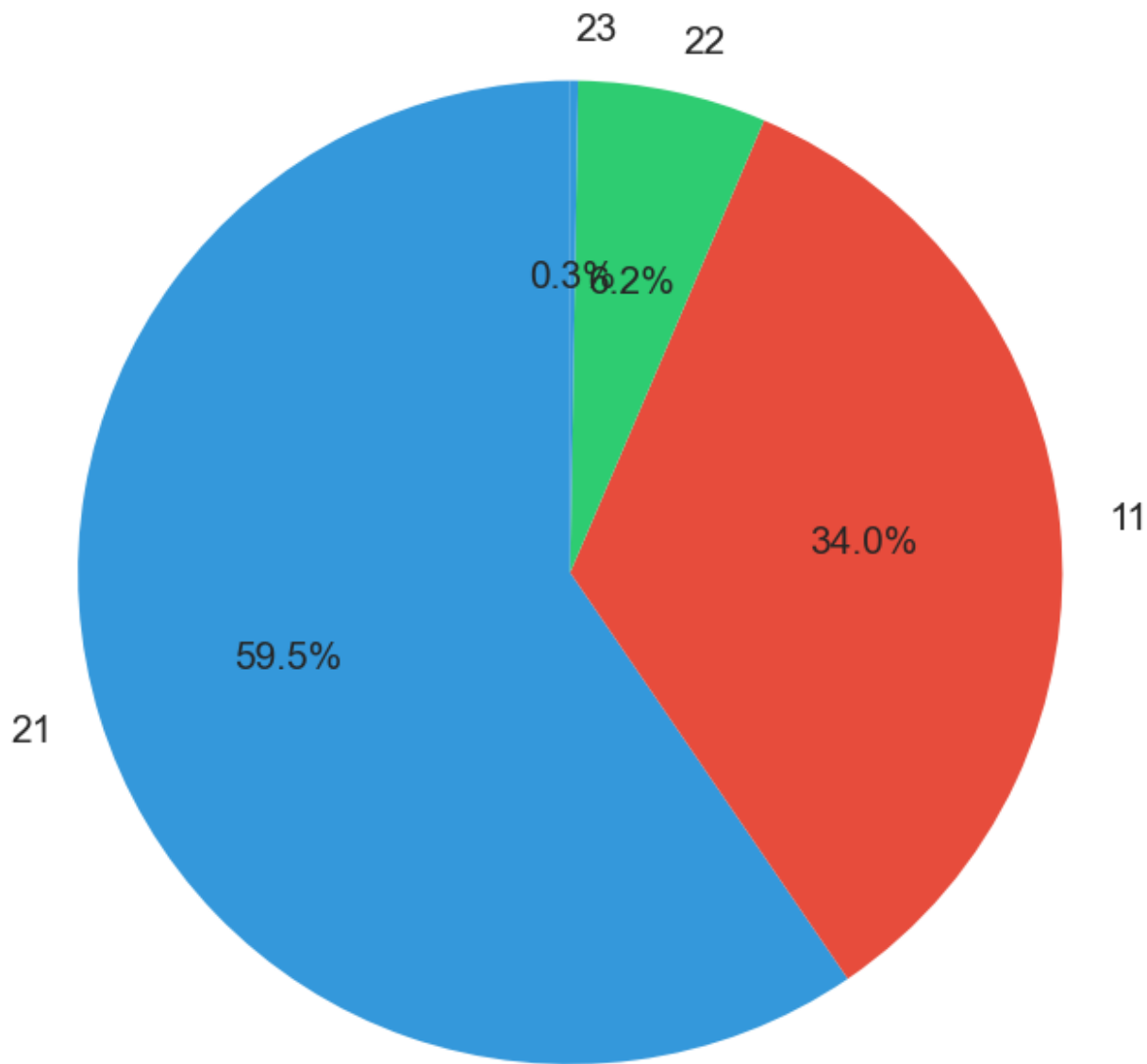
HCPCS	Description	Count
99291	Critical Care, Initial First Hour	68
99222	Initial Hospital Care, Level 2	30
99233	Subsequent Hospital Care, Level 3	45 (combined)
99223	Initial Hospital Care, Level 3	24
99213	Office/Outpatient Visit, Level 3	34 (combined)
90833	Psychotherapy with E&M, 30 min	16
92557	Comprehensive Audiometry	14
99442	Telephone Services, 11-20 min	13

Interpretation: The procedure mix shows:

- **High acuity care:** Critical care (99291) is the most common procedure, aligning with the respiratory failure diagnoses
- **Inpatient E&M:** Initial and subsequent hospital care codes dominate, consistent with the "INPATIENT" place of service majority
- **Specialty services:** Psychotherapy and audiometry indicate diverse specialty care
- **Telehealth:** Telephone services represent modern care delivery models

Question 5: Place of Service Analysis

Q5: Claims by Place of Service



Place of Service	Claims	Percentage
21 - Inpatient Hospital	231	59.5%
11 - Doctor's Office	132	34.0%
22 - Outpatient Hospital	24	6.2%
23 - Emergency Room	1	0.3%

Interpretation: Nearly 60% of claims are for inpatient hospital services, consistent with the critical care and surgical provider mix. About one-third are office visits, representing outpatient follow-ups and routine care. The low ER claim count (0.3%) suggests this sample focuses on scheduled/admitted care rather than emergency services.

Part 3: Advanced Analysis with Joins

Question 6: Claims with High Service Line Counts (5+ Lines)

Claim ID	Provider	Lines	Total Charges
36794825	SB CHILDREN'S SERVICE	7	\$1,163
36668119	UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	6	\$1,030
36740402	UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	6	\$945
36710175	UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	5	\$873
36757684	UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	5	\$873

Interpretation: Only 5 claims (1.3%) have 5 or more service lines. OB/GYN accounts for 4 of these, likely representing comprehensive prenatal or postpartum visits that include multiple bundled services. The Children's Service claim with 7 lines suggests a complex pediatric encounter.

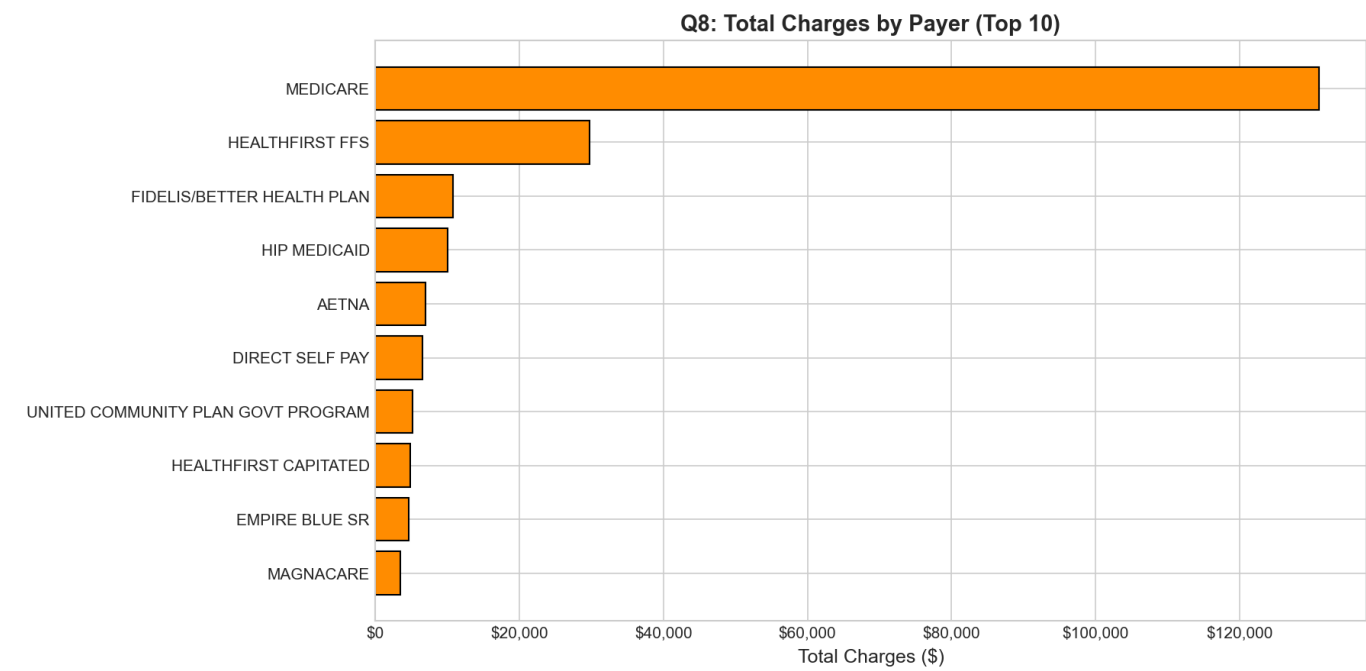
Question 7: Diagnosis Codes Associated with CPT 99291 (Critical Care)

ICD-10 Code	Description	Frequency
J96.01	Acute respiratory failure with hypoxia	53
G93.5	Compression of brain	34
E78.5	Hyperlipidemia, unspecified	33
I61.9	Intracerebral hemorrhage	26
D64.9	Anemia, unspecified	25

Interpretation: Critical care services (99291) are most commonly associated with:

- **Respiratory failure** (J96.01) - the primary reason for critical care in 78% of 99291 claims
- **Neurological emergencies** - brain compression and hemorrhage codes reflect the neurosurgical patient population
- **Chronic comorbidities** - hyperlipidemia and anemia commonly appear as secondary diagnoses on critical care claims

Question 8: Total Charges by Payer



Payer	Total Charges	Avg Charge/Claim	Claims
MEDICARE	\$131,008	\$437	242
HEALTHFIRST FFS	\$29,794	\$408	46
FIDELIS/BETTER HEALTH PLAN	\$10,810	\$292	26
HIP MEDICAID	\$10,014	\$223	17
AETNA	\$6,930	\$1,155	6
DIRECT SELF PAY	\$6,575	\$1,096	6
UNITED COMMUNITY PLAN	\$5,175	\$862	5

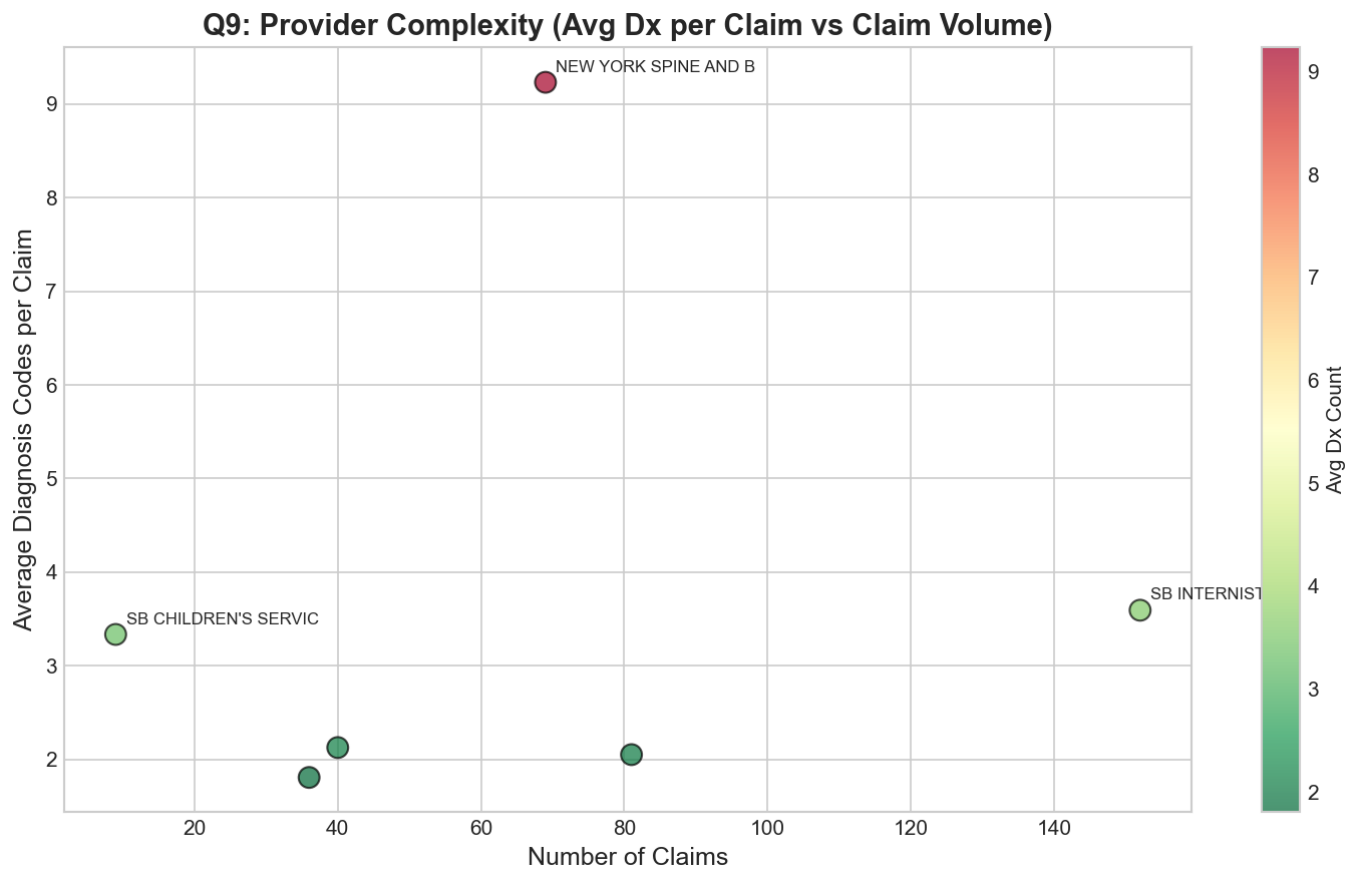
Interpretation:

- **Medicare** generates the most revenue (\$131K) due to volume
- **AETNA and Direct Self Pay** have the highest average charges per claim (~\$1,100), suggesting commercial/self-pay patients may have more complex or higher-reimbursement services
- **Medicaid payers** have lower average charges (\$200-400), likely reflecting lower fee schedules and service mix

Part 4: Creative Analysis

Question 9: Provider Complexity Analysis

Research Question: Which providers bill for the most complex cases (measured by average diagnosis codes per claim)?



Provider	Avg Dx Codes	Claim Count
NEW YORK SPINE AND BRAIN SURGERY	9.23	69
SB INTERNISTS	3.59	152
SB CHILDREN'S SERVICE	3.33	9
UNIV.ASSOC.IN OBSTETRICS&GYNECOLOGY	2.13	40
SB SURGICAL ASSOCIATES	2.05	81
SB PSYCHIATRIC ASSOCIATES	1.81	36

Interpretation:

New York Spine and Brain Surgery stands out dramatically with an average of **9.23 diagnosis codes per claim** - more than double the next highest provider. This reflects:

- 1. **High case complexity:** Neurosurgical patients typically have multiple comorbidities
- 2. **Critical care overlap:** Many spine/brain surgery patients likely require ICU-level care
- 3. **Thorough documentation:** Complex procedures require comprehensive diagnosis coding for medical necessity

In contrast, Psychiatric Associates has the lowest complexity (1.81 dx/claim), which is expected since mental health visits typically focus on primary psychiatric diagnoses with fewer medical comorbidities documented.

Recommendation: The compliance team should focus documentation audits on high-complexity providers to ensure all diagnosis codes are properly supported.

Summary of Key Findings

1. **Medicare dominates** the payer mix at 62%, with Medicaid-related plans accounting for 25%
2. **Critical care** (99291) is the most common procedure, associated primarily with respiratory failure
3. **Neurosurgical services** drive the highest case complexity (9+ diagnoses per claim)
4. **Inpatient services** represent 60% of claims, with the remaining 40% split between office and outpatient settings
5. **SB Internists** generates the highest claim volume but moderate complexity, typical for internal medicine

Analysis conducted using Python with pandas, matplotlib, and seaborn.